

Yifan Wang

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• EDUCATION

06/2022 – Present Nanjing, China

PHD Southeast University, School of Automation

Thesis: *Distributed Optimization and Learning under Constrained Information Exchange*

Supervisor: Prof. Xianghui Cao

08/2025 – 10/2026 Padua, Italy

VISITING PHD STUDENT University of Padua, Department of Information Engineering

Research Topics:

Communication-Efficient Interior-Point Methods for Model Predictive Control

Adaptive Step-Size Methods for Distributed Optimization

Host: Prof. Luca Schenato

09/2019 – 06/2022 Jinan, China

MASTER'S DEGREE Shandong University, School of Control Science and Engineering

Thesis: *Distributed Energy Management and Optimization in Smart Grids*

Supervisor: Prof. Shuai Liu

09/2015 – 06/2019 Daqing, China

BACHELOR'S DEGREE Northeast Petroleum University, School of Electrical Information Engineering

Thesis: *Consensus-Based Formation Control of UAV Swarms*

• RESEARCH INTERESTS

- Communication-efficient algorithm design in distributed optimization.
- Fundamental trade-offs in privacy-preserving federated learning.
- Applications of distributed algorithms in power systems and real-time MPC problem.

• PUBLICATIONS

Journal articles

1. **Y. Wang**, X. Cao, S. Jin and M.-Y. Chow, "A Novel Privacy Enhancement Scheme With Dynamic Quantization for Federated Learning," *IEEE Transactions on Signal Processing*, 2026.
2. **Y. Wang**, C. Zhang and X. Cao, "Resilient Distributed Optimization With Trust Opinion Dynamics for Cyber-Physical Systems over Directed Graphs," *IEEE/CAA Journal of Automatica Sinica*, 2026.
3. **Y. Wang**, S. Liu, X. Cao and M.-Y. Chow, "An Operator Splitting Scheme for Distributed Optimal Load-Side Frequency Control With Nonsmooth Cost Functions," *IEEE Transactions on Automatic Control*, 2024.
4. **Y. Wang**, S. Liu, B. Sun and X. Li, "A Distributed Proximal Primal-Dual Algorithm for Energy Management With Transmission Losses in Smart Grid," *IEEE Transactions on Industrial Informatics*, 2022.

Conference papers

1. **Y. Wang**, L. Ballotta, R. Carli, X. Cao and L. Schenato, "Pursuing Optimal Stepsize in Adaptive Gradient-Based Quadratic Optimization," *IFAC-PapersOnLine*, 2026.
2. **Y. Wang** and S. Liu, "A Distributed Proximal-Based Primal-Dual Algorithm for Composite Optimization With Coupled Constraints," *International Conference on Control & Automation*, 2022.
3. **Y. Wang**, S. Han and S. Liu, "Distributed Consensus-Based Algorithm for Dynamic Economic Dispatch With Wind Turbine and Energy Storage," *International Conference on Information, Cybernetics, and Computational Social Systems*, 2020.

Under review

1. **Y. Wang**, L. Ballotta, R. Carli, A. Iannelli, X. Cao and L. Schenato, "Adaptive Stepsizes With Certified Convergence in Distributed Gradient Tracking With Quadratic Costs," 2026. [Submitted to *IEEE Transactions on Automatic Control*].
2. **Y. Wang**, M. Wang and X. Cao, "Noise-Robust Distributed Optimization Over Directed Graphs With Row Stochastic Matrices," 2026. [Submitted to *IEEE Transactions on Control of Network Systems*].
3. **Y. Wang**, S. Jin and X. Cao, "Privacy-Preserving Online Federated Learning under Age-Dependent Differential Privacy," 2026. [Submitted to *IEEE Transactions on Signal Processing*].

4. X. Gu, Z. Guo, **Y. Wang** and S. Liu, “Prescribed Performance-Triggered Non-Singular Practical Prescribed-Time Control,” 2026. [Submitted to IEEE Transactions on Cybernetics].
5. Q. Xie, **Y. Wang** and X. Cao, “Byzantine-Resilient Distributed State Estimation With Local Differential Privacy: A Visible-Hidden State Decomposition Approach,” 2026. [Submitted to IEEE Transactions on Information Forensics and Security].

• LANGUAGE SKILLS

Languages:

• HONORS AND AWARDS

Young Elite Scientists Sponsorship Program by CAST for Doctoral Students, 2024

Awardee; Funding: CNY 40,000.

Southeast University Postgraduate Innovation Capacity Promotion Program, 2024

Principal Investigator; Funding: CNY 48,000.

Postgraduate Research & Practice Innovation Program of Jiangsu Province, 2023

Principal Investigator; Funding: CNY 15,000.

• SERVICE

Reviewer activity

- IEEE Transactions on Automatic Control
- IEEE Journal on Selected Areas in Communications
- IEEE Transactions on Signal and Information Processing over Networks
- IEEE Transactions on Industrial Informatics
- IEEE/CAA Journal of Automatica Sinica
- IEEE Transactions on Control of Network Systems
- ISA Transactions